NAKUL-Med: Spectral-Graph State Space Models with Dynamics Kernels for Medical Signals

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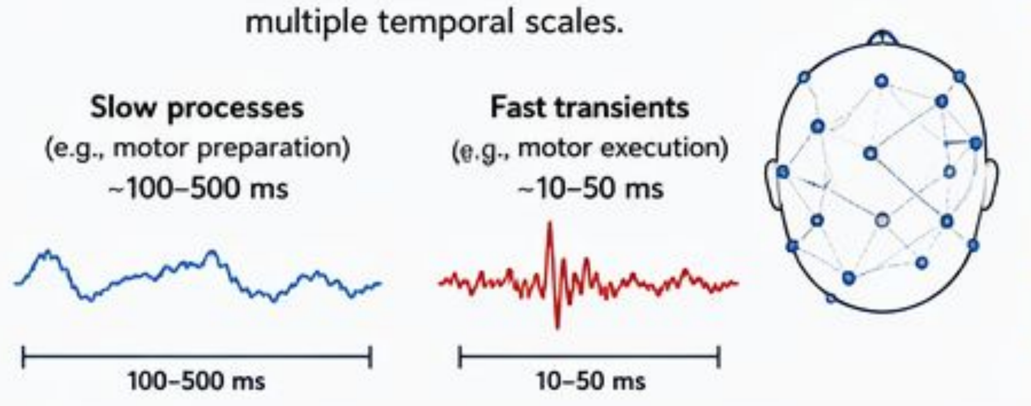
MOTIVATION

Medical signals are multi-scale, long-range, and spatially structured—demanding models that adapt in time, frequency, and space.

1. Multi-Scale Temporal Dynamics

EEG/MEG/BCI signals exhibit events at multiple temporal scales.



2. Long-Range Dependencies

Neural oscillations (alpha, beta, etc.) are periodic and span long time horizons.

Distant timesteps are semantically related but hard to access with local/state updates.

3. Spatial Structure Matters

Neural activity follows brain topology. Electrodes form a fixed geometric graph with meaningful spatial relationships.

We need a model that

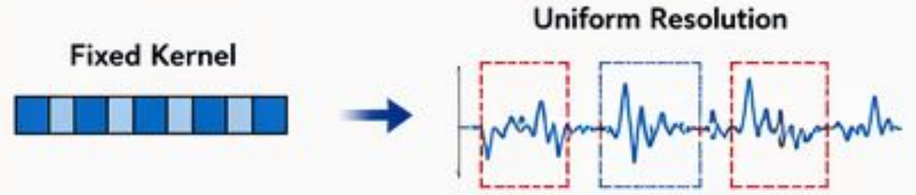
- Adapts to multiple temporal scales
- Captures global dependencies efficiently
- Leverages spatial structure with inductive biases

PROBLEM STATEMENT

Limitations of existing models for medical time series (e.g., EEG, MEG, BCI).

1. Fixed Temporal Scales (SSMs / CNNs)

Standard SSMs use fixed convolutional kernels, enforcing uniform temporal resolution.

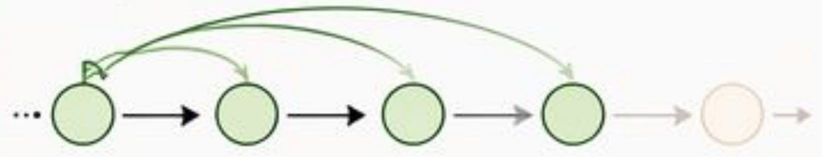


Miss either slow trends or fast transients.

✕ Cannot adapt kernel temporal scales per input.

2. Limited Global Context (SSMs)

Markovian state updates depend only on the previous state → exponentially decaying receptive field.

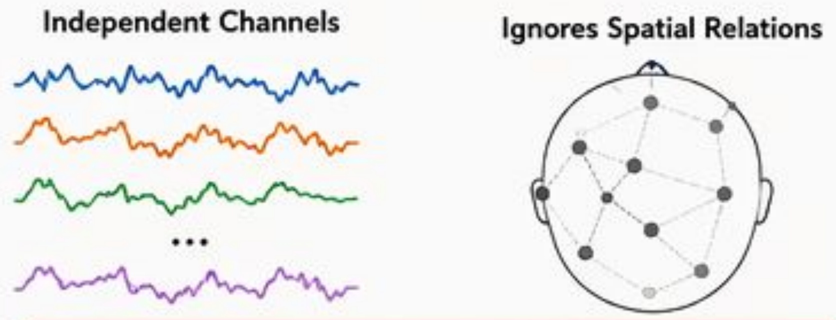


Struggles with long-range periodic patterns (e.g., alpha/beta oscillations in EEG).

✕ Difficult to capture long-range dependencies efficiently.

3. No Spatial Structure (SSMs)

SSMs process each channel independently, ignoring electrode geometry.



✕ Misses spatial patterns across brain regions.

Takeaway

Existing models fail to simultaneously address multi-scale temporal adaptation, global context, and spatial structure.

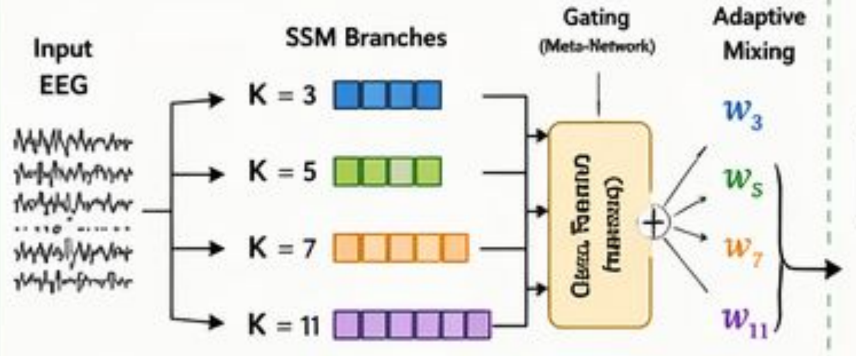
We need a unified model that integrates all three.

OUR CONTRIBUTIONS (NAKUL-Med)

Spectral-Graph State Space Model with Dynamic Kernels

1. Dynamic Kernel Selection

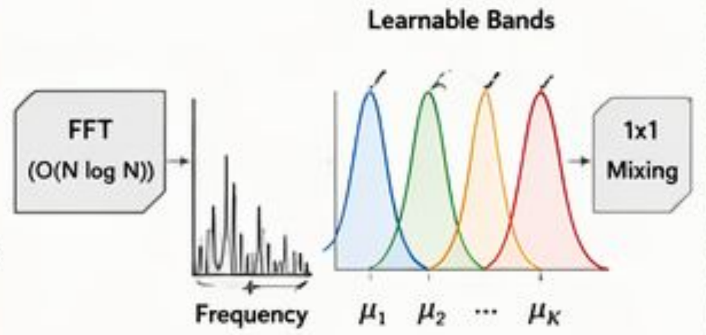
Parallel SSM branches with multiple kernel sizes (3, 5, 7, 11 timesteps; 12–44 ms @ 250Hz).



✓ Adapts temporal scale per input by predicting mixing weights (not just kernel weights).

2. Learnable Spectral Bands

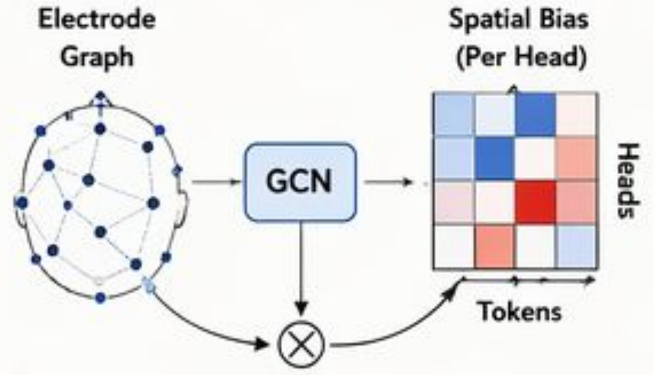
FFT-based mixing with learnable Gaussian band filters (centers μ_K, widths σ_K).



✓ Discovers task-relevant frequency ranges instead of using fixed bands (e.g., alpha, beta).

3. Graph-Guided Spatial Biases

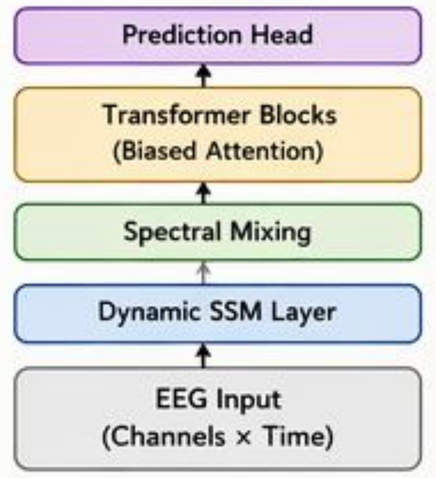
Use fixed electrode topology via GCN to produce spatial biases for multi-head attention.



✓ Injects geometric priors into attention while preserving flexible, data-driven connectivity.

4. Unified Architecture

Seamlessly integrates temporal, spectral, and spatial modeling.



Why it Matters

- ✓ Adapts to multi-scale dynamics
- ✓ Captures long-range dependencies efficiently
- ✓ Leverages brain topology with inductive biases
- ✓ Designed for medical time series (EEG/MEG/BCI)

NAKUL-Med sets a new state-of-the-art on multiple EEG benchmarks with superior generalization across datasets and subjects.

NAKUL-Med: State-of-the-Art Performance on Multiple Medical Benchmarks

1 LEADERBOARD: Best Overall Performance

NAKUL-Med

89.3% AVG ACCURACY across 5 datasets

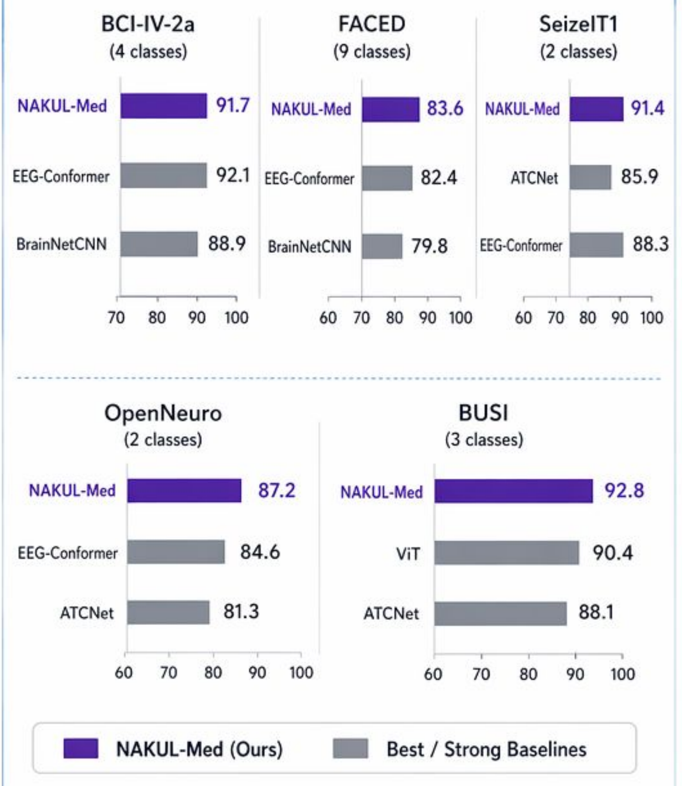
2.5M PARAMETERS — Highest avg accuracy with compact model size

Top Competitors (by Avg Accuracy)

2	EEG-Conformer	86.9%
3	ATCNet	84.5%
4	BrainNetCNN	84.4%
5	Mamba-2	83.4%

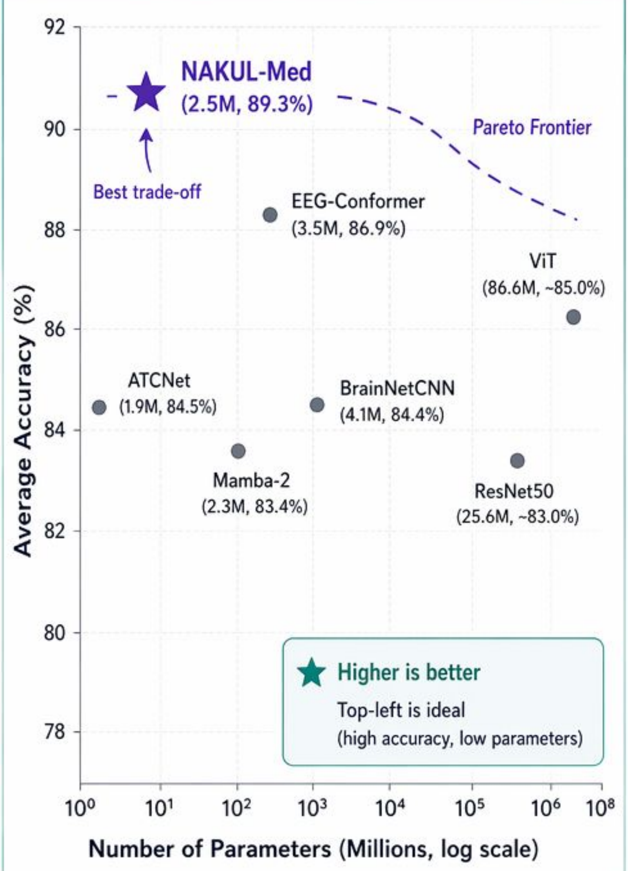
★ Outperforms CNN, RNN, SSM, Transformer & Hybrid models

2 DATASET-WISE PERFORMANCE (Accuracy %)



✓ NAKUL-Med achieves top or near-top performance on all 5 datasets

3 ACCURACY vs MODEL EFFICIENCY (Avg Accuracy across 5 datasets)



BEST AVERAGE ACCURACY — 89.3% across 5 datasets

PARAMETER-EFFICIENT — Only 2.5M parameters

CONSISTENT & ROBUST — Top performance across diverse tasks and datasets

NAKUL-Med — Sets a new state-of-the-art!

SCAN ME

Efficiency Comparison of NAKUL with State-of-the-Art Models

	Method	Params (M)	FLOPs (G)	Memory Bandwidth (GB/s)	Energy (mJ/sample)	Params/Ace Efficiency	FLOPs/Ace Efficiency	Energy/Ace Efficiency
CNN-based	EEGNet [19]	0.003	0.042	2.1	0.8	0.03	0.48	0.9
	ResNet50 [28]	25.6	4.1	18.4	12.3	288.6	46.2	138.7
	3D-ResNet [14]	33.2	5.8	24.7	17.6	393.8	68.8	208.9
Transformer-based	ViT [10]	86.6	17.6	42.3	28.4	957.7	194.7	314.1
	EEG-Conformer [29]	3.5	2.8	12.6	8.7	40.3	32.2	100.1
	BrainNetCNN [18]	4.1	1.9	10.8	7.4	48.6	22.5	87.7
SSM-based	Vanilla Mamba [11]	2.1	0.87	4.3	2.1	25.7	10.6	25.7
	Mamba-2 [5]	2.3	0.94	4.8	2.4	27.6	11.3	28.8
	NAKUL (Ours)	**2.5**	**1.43**	**6.2**	**3.8**	**28.0**	**16.0**	**42.6**

How much more efficient is NAKUL? (vs. selected strong baselines)

	Params (M)	FLOPs (G)	Memory Bandwidth (GB/s)	Energy (mJ/sample)	Params/Ace Efficiency	FLOPs/Ace Efficiency	Energy/Ace Efficiency
vs. Conformer	1.4× fewer	2.0× fewer	2.0× lower	2.3× lower	1.4× better	2.0× better	2.3× better
vs. ViT	34.6× fewer	12.3× fewer	6.8× lower	7.5× lower	34.2× better	12.2× better	7.4× better
vs. Vanilla Mamba	1.2× more	1.6× more	1.4× higher	1.8× higher	1.1× worse	1.5× worse	1.7× worse

★ NAKUL achieves strong efficiency across all key metrics, outperforming state-of-the-art baselines with fewer parameters, lower computation, lower memory access and lower energy consumption.

NAKUL Ablation Study: Impact of Components and Combinations

Configuration		BCI-IV-2a (Acc %)	Δ (vs Full)	FACED (Acc %)	Δ (vs Full)
Full Model (NAKUL)		**91.7**	–	**83.6**	–
Individual Component Removal	w/o NeuraSpectraNet	88.9	-2.8	80.4	-3.2
	w/o Dynamic Kernels	89.2	-2.5	81.1	-2.5
	w/o Graph Mixing	89.4	-2.3	81.3	-2.3
Pairwise Combinations	NeuraSpectra + Dynamic	90.1	-1.6	82.3	-1.3
	NeuraSpectra + Graph	90.3	-1.4	82.6	-1.0
	Dynamic + Graph	89.7	-2.0	81.9	-1.7
Single Component Only	Only NeuraSpectraNet	87.8	-3.9	78.2	-5.4
	Only Dynamic Kernels	86.7	-5.0	77.4	-6.2
	Only Graph Mixing	86.4	-5.3	76.9	-6.7
Vanilla Mamba Baseline [11]		84.2	-7.5	74.8	-8.8

All three components are essential. Removing any component or using partial combinations consistently degrades performance across both datasets.

Best Accuracy 91.7 / 83.6 — Largest Gain up to -8.8% — Synergy Works All 3 > Any Subset

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